Challenge Problems for Midterm Preparation

Regression – Analytical Questions

1. Calculate the standard error of the regression.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R						
R Square	0.80					
Adjusted R Square						
Standard Error						
Observations						
ANOVA						
	df	*SS*	*MS*			
Regression			350			
Residual						
Total	400					
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept						
Excess Market Return						

2. Conduct a one-sided hypothesis test to test of the y-intercept is greater than 0.50. What is the P-value of the test?

SUMMARY OUTPUT						
Regression Statistics						
Multiple R						
R Square						
Adjusted R Square						
Standard Error						
Observations	400					
ANOVA						
	df	*SS*	*MS*			
Regression						
Residual						
Total						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	0.30		0.50			
Excess Market Return						

3. What is the 80% confidence interval for the slope?

SUMMARY OUTPUT						
Regression Statistics						
Multiple R						
R Square						
Adjusted R Square						
Standard Error						
Observations						
ANOVA						
	df	*SS*	*MS*			
Regression						
Residual	400					
Total						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept						
Excess Market Return	0.80	0.20				

4. Apple and the market have risk premiums of 10% and 5%. Below is the regression output from regressing Apple's excess returns onto the market's excess returns. What is the variance of the market?

SUMMARY OUTPUT						
Regression Statistics						
Multiple R						
R Square						
Adjusted R Square						
Standard Error						
Observations						
ANOVA						
	df	*SS*	*MS*			
Regression						
Residual	400					
Total						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept			3.00		1.01	
Excess Market Return						

5. Conduct a two-sided hypothesis test to test if the slope is different from 0.9. Use significance level of 10%.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R						
R Square						
Adjusted R Square						
Standard Error						
Observations						
ANOVA						
	df	*SS*	*MS*			
Regression		1000				
Residual			10			
Total		6000				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept						
Excess Market Return	1.40	0.70				